CIS 635: Assignment 2a

Supervised Learning I: Regression with Linear Regression and k-Nearest Neighbors

Deadline: Please check the Blackboard post **Total points:** 5

Objective

This assignment provides hands-on practice with **regression**, the supervised task of predicting a numeric target. You will build and compare two very different models: **linear regression**, a parametric model that is easy to interpret, and **k-nearest neighbors (kNN) regression**, a non-parametric, distance-based model. You will also practice **hyperparameter optimization** with cross-validation and see why feature scaling matters for distance-based methods. The emphasis is on sound evaluation: comparing against a baseline, using a held-out test set, and avoiding data leakage. The data are samples; do not make global conclusions from these samples.

Data

Use the following file:

- **Medical insurance charges:** `https://raw.githubusercontent.com/stedy/Machine-Learning-with-R-datasets/master/insurance.csv`
  Target: `charges` (individual annual medical costs, USD). Features: `age`, `sex`, `bmi`, `children`, `smoker`, `region`.

General rules. Hold out a test set (e.g., 80/20) *before* any tuning, and use it only for the final evaluation. Fix a random seed so your results are reproducible. Any scaling or encoding that learns from the data must be fit on the training data only (a `Pipeline` is the easiest way to guarantee this, including inside cross-validation).

Tasks

(a) Data Preparation and EDA — 1 point

(i) Check for missing values and duplicate records and handle them. Encode the categorical features (`sex`, `smoker`, `region`) and briefly justify your encoding. (0.5)

(ii) Describe the distribution of `charges` (e.g., with a histogram), and create one plot showing how `charges` relates to `age` or `bmi`, colored by `smoker`. What do you expect each model to find easy or hard? (0.5)

(b) Linear Regression — 1 point

(i) Fit a **linear regression** model. Report test-set R^2, MAE, and RMSE, and compare them with a naive baseline that always predicts the training-set mean of `charges`. (0.5)

(ii) Interpret the coefficients of `smoker` and `bmi` in plain language (units included). (0.5)

(c) k-Nearest Neighbors Regression and Hyperparameter Optimization — 2 points

(i) Fit a kNN regressor with the default $k = 5$ *twice*: once on the unscaled features and once with standardized features. Report the test-set metrics for both and explain the difference. (0.5)

(ii) Using a pipeline (scaler + kNN), optimize the hyperparameters with **5-fold cross-validation** on the training data only. Search at least `n_neighbors` $\in \{1, 2, \ldots, 30\}$ and `weights` $\in \{$`uniform`, `distance`$\}$ (you may add others, e.g., the distance metric). Report the best hyperparameters and their cross-validated RMSE, then report the test-set R^2, MAE, and RMSE. (1)

(iii) Plot the cross-validated RMSE against k. Explain what happens at very small and very large k in terms of the bias–variance trade-off (under- vs. overfitting). (0.5)

Optional: You may also tune a regularized linear model (Ridge or Lasso, tuning `alpha`) for comparison. This is not required.

(d) Model Comparison — 1 point

(i) Summarize the baseline, linear regression, and tuned kNN in one table of test-set metrics. Create a residual plot (residuals vs. predicted values) for each model. (0.5)

(ii) Which model would you choose, and why? Discuss the trade-off between accuracy and interpretability, and what the residual plots reveal that the metrics alone do not. (0.5)

Submission Guidelines

Submit a runnable notebook (Python, R, or another programming language of your choice) through Blackboard (the link of your notebook, which stays in your Google drive or else where, is not accepted). Include sufficient markdown/comments to explain your analysis and decisions. Use appropriate labels, titles, units, and legends in visualizations. Report numerical results to three decimal places where appropriate, and report the random seed(s) you used.

Important: The goal is not to find the single highest score. Focus on a sound modeling process: use **proper train/test separation, cross-validation, appropriate metrics, and clear interpretation** to support your conclusions.